ZICHEN ZHU

Berkeley, CA | zichen_zhu@berkeley.edu | github.com/zhuzccn-rgb

EDUCATION

University of California, Berkeley

First-year undergraduate | Conservation and Resource Studies

TECHNICAL PROJECTS

ONE Mind / SEGMA | Framework Design, Prototype

- Conceived a visual-memory system for smart glasses and designed its modular framework; current scope is a 2D image-sequence prototype.
- Prototype combines YOLO detection, center-weighted attention, nearest-neighbor ID matching, and time-decaying state. Long-term memory remains unfinished; repository is private.

AI-examiner | Concept & Workflow Design, Prototype

- Designed the workflow for extracting questions from PDFs and assembling exams by topic and score; prototype code uses LangGraph, OCR, and knowledge retrieval.

Folio / pdf2epub | Project Concept, Prototype

- Originated a local conversion tool for Manning-style technical PDFs; prototype includes chapter navigation, code-block and image extraction, and a Flask interface with live progress.

Local AI Inference Server | Self-directed Team Project

- Initiated a shared AI tutoring server and personally assembled it with peers using repurposed hardware; coordinated technical planning, environment setup, and troubleshooting records.
- Invited student feedback and presented the project at a club showcase, balancing hardware cost and accessible AI tutoring.

MoS2 Semiconductor Research Experience | Summer 2024

- Participated in supervised sample preparation and experimental documentation; discussed synthesis conditions and adjustments with a mentor.

LEADERSHIP AND COMMUNITY

Science and Technology Club | President, High School

- Organized technology projects and demonstrations, including an ESP32-based automatic bicycle gear-shifting concept and prototype work.

Global Future Space Scholars Meet | Team Leadership & Presentation

- Coordinated a multidisciplinary space-city design proposal and final presentation; received Overall Best Speaker recognition.

Student Union | President, High School

- Helped organize major campus events, including a Christmas gala and Halloween activities; coordinated student participation and event execution.

SKILLS AND INTERESTS

Skills: PC/server component selection and assembly; local LLM deployment; software-environment setup and troubleshooting; project workflow design and technical presentations.

Interests: visual memory for wearable AI; OCR and retrieval for study tools; low-cost inference on repurposed hardware; the trade-offs between computing performance, energy use, and accessibility.